Halt-Failure Behaves Like a Class-Structured Residual-Stream Attractor in Fine-Tuned OCR Vision-Language Models

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Abstract

Fine-tuned OCR vision-language models (OCR-VLMs) sometimes fail to stop: instead of emitting end-of-sequence (EOS), decoding runs to the generation cap and surfaces as phantom table rows, runaway loops, and hallucinated structural markers. Studying this *halt-failure* in a 3B-parameter OCR fine-tune, we advance the *hypothesis* that it is not a single broken layer but behaves like a stable, class-structured attractor in the residual stream. Three findings motivate this reading. (i) Per-class halt directions are linearly decodable across the decoder (AUC 0.876 to 0.983) and content-decoupled under a pre-registered test. (ii) The network computes a should-halt signal only late: the logit-lens EOS gap is most negative at layer 24 and flips positive by layer 32, and EOS is suppressed only relatively, out-scored by continuation tokens by roughly 20 logits. (iii) Six converging perturbation nulls — spanning single-direction project-out, component-resolved patching, induction-head zeroing, prompt-engineering counterfactuals, SAE-feature ablation, and multi-layer coordinated patching — all return the pre-registered FAIL/null verdict (N ranging 2 to 22 documents): no single-direction or single-layer perturbation changes the token count. The separable geometry is fine-tune-introduced: a matched base-model comparison on the same Qwen2.5-VL-3B backbone shows no such signature even on the base model's own cap-hit documents (positive-versus-control absolute Cohen's-d ratio 1.12 / 0.78 / 0.92 at layers 16 / 20 / 24, all near 1× and below the pre-registered 2.0× bar), whereas the fine-tune's ratios are 3.79 / 2.74 / 2.78, so fine-tuning introduces and sharply amplifies the structure. We are explicit that these nulls do not settle the mechanism: all single- and multi-site perturbations we ran returned null at this protocol's power, and whether the mechanism is genuinely distributed or the protocol is under-powered at the halt-relevant locus is *not resolved* by these data. A norm-scaled positive control (3 cap-hit positives, none of which escapes the cap at any scale) rules out only the strongest version of the under-powered objection (the protocol is not inert: it visibly alters content), leaves the weaker version open, and is further blunted by a clean-halt control in the same run whose own output collapsed under the perturbation (a norm-shock signature). The halt-failure surface symptom is not unique to the studied fine-tune: under a strict stop-reason criterion it reproduces on 10 of 16 doc-model cells across four model families, but the matrix is small and the aggregate verdict abstains, so we make no strong generality claim. We close with an honest production trade-off, a runtime monitor that cuts runaway length 37.68% at zero control false-positives, and show via grey-noise image inpainting that image content is sufficient to sustain the loop (all 5 of 5 named positives collapse to short EOS halts).

AI assistance disclosure. This preprint was produced with substantial AI assistance: an AI research agent drafted text, ran and logged experiments, and assembled the manuscript under the author's direction. The author specified the research questions, designed and pre-registered the falsification criteria, audited every numeric claim against on-disk results files, and made all interpretive and editorial decisions. Numeric claims that could not be reproduced from a results file were marked unverified and excluded from the load-bearing argument rather than retained. The author takes full responsibility for the content, including any remaining errors.

1 Introduction

Autoregressive vision-language models for document understanding occasionally enter a state from which they never emit EOS. On a structured document the decoder will, for example, keep emitting `<tr><td></td></tr>` phantom rows, repeat a single filled cell value hundreds of times, or cycle a LaTeX command indefinitely, until the generation harness force-stops it at the token budget. From a production standpoint this is a reliability bug. From a mechanistic standpoint it is an unusually clean window into how a fine-tuned model represents the *decision to terminate* and why that decision can be silently overridden.

This paper stakes a single, deliberately-hedged thesis: in fine-tuned OCR-VLMs, halt-failure *behaves like* a distributed residual-stream attractor rather than a single-layer bug. We label this an attractor *hypothesis*, not a settled mechanism. We do not claim to have found "the halt circuit." On the contrary, the core of our causal evidence is a series of perturbation experiments that *fail* to move behavior when we intervene at any single direction or layer. We treat those nulls as first-class results, but we are equally explicit about what they cannot show: a converging set of nulls at a fixed protocol power is consistent with a genuinely distributed mechanism *and* with an under-powered protocol that cannot perturb the halt-relevant direction at the halt-relevant locus. The present data do not distinguish these, so we report the attractor reading as the framing that fits the full evidence pattern, not as a mechanism we have isolated (Section 5, Section 9).

The two payoff results that set up the rest of the paper are (i) that per-class halt directions are linearly decodable and content-decoupled across the decoder (Section 4), and (ii) that the network computes a should-halt signal late in the stack that arrives too late to override an already-committed continuation state (Section 4). Everything else either characterizes the phenomenon (Section 2), establishes the method (Section 3), or stress-tests the attractor reading causally (Section 5).

2 The phenomenon and corpus

2.1 Failure taxonomy

Halt-failure is not monolithic. On real documents it surfaces as several distinct *surface classes*: filled-cell repetition (a single cell value repeated), bare-word repetition, LaTeX math or command loops, LaTeX structure loops, HTML phantom rows, tag spam, punctuation runs, and constant-value repetition, among others. These are surface symptoms of a shared underlying failure to terminate. One of our central method choices (Section 3) is to probe for halt directions *per class* rather than assume one shared representation.

2.2 Corpus construction and a non-determinism disclosure

Real public documents trigger runaway decoding. Our review corpus is drawn from 1,139 inference runs over 424 unique pages, yielding a set of cap-hit document IDs that resolve to a confirmed positive set. After excluding boundary cases, false positives, and unclassified rows, the corpus comprises 56 confirmed infinite-generation positives spanning 12 populated surface classes. We report the on-disk figure deliberately, rather than a higher headline count that does not reproduce from the underlying corpus index.

We also disclose a device-induced shrink in the stable positive set. When we re-screened a 123-document set on a CUDA GPU rather than on an Apple-silicon (MPS) backend, the strict CUDA-positive hit rate was 6 of 123, or 4.88%. Of the MPS-positive disagreements, 8 flipped to CUDA-control, and 0 new CUDA-only positives appeared. In other words, the cross-backend non-determinism shrank the stable positive set rather than enlarging it; we treat the smaller, device-consistent CUDA set as the conservative ground truth.

3 Method

We use three complementary instruments, all pre-registered with falsification criteria before the validating runs.

Per-class halt-direction probing. For each surface class and each of 10 sampled decoder layers (L0, L4, L8, L12, L16, L20, L24, L28, L32, L35), we fit a linear probe to separate halt-state residual activations from non-halt activations, evaluated under leave-one-position-out (LOPO) and leave-one-document-out (LODO) cross-validation. We report per-class AUC, a pooled global AUC, and pairwise cosine similarities between per-class directions to test content decoupling.

Logit-lens triangulation. At every cached layer we apply the final RMSNorm and the unembedding to the residual stream and read off the EOS-token logit, comparing cap-hit positives against clean controls. This localizes *where in depth* a halt signal becomes legible at the unembedding [6].

Block-contribution decomposition. We decompose per-layer residual updates into block contributions and measure the cosine alignment between a layer's block output and the per-class halt direction, separating where a halt representation is *built* versus merely *read* [19].

Pre-registered falsification. Each causal test in Section 5 carries a numeric PASS threshold fixed before the run (for example, the L0 sufficiency test pre-registers a ≥ 40 percentage-point cap-hit increase as the PASS criterion). Significance for the cross-document probe uses a block-shuffle null at multiple block sizes ($B = 5, 10, 20$) with Bonferroni correction, the conservative $B = 20$ block size being the load-bearing one for autocorrelation. We ran $N_{\text{perm}} = 300$ permutations, so the reported $p = 0.0033$ is the resolution floor $1/(N_{\text{perm}} + 1) = 1/301$: it records that the observed

AUC exceeded every one of the 300 block-shuffled permutations, an exceedance count of 0, not a finer measured separation. We therefore read it as "$p \leq 0.0033$" rather than an exact value, and note that a tighter floor below the 0.005 Bonferroni threshold would require a re-run at larger N_{perm}. Under this conservative $B = 20$ null only layers 20 and 24 clear the Bonferroni threshold of 0.005 across the ten probed layers.

4 Mechanism findings

4.1 Halt representations are class-specific and content-decoupled

Per-class halt directions are strongly linearly decodable across the whole decoder, consistent with the linear representation hypothesis that high-level concepts are encoded as directions in the residual stream [26]. Across all (layer, class) pairs the per-class AUC ranges from 0.876 to 0.983 (minimum at filled-cell-repeat layer 8 = 0.8763, maximum at filled-cell-repeat layer 24 = 0.9826). Critically, the three classes peak at *different* layers: LaTeX peaks at layer 8 = 0.9785, filled at layer 24 = 0.9826, and bare-word at layer 20 = 0.9173 (Figure 1). Distinct peak layers are direct evidence that these are class-specific representations rather than one shared halt direction, matching the categorical/hierarchical concept-geometry picture in which related concepts occupy structured, separable subspaces [27].

The pre-registered structural-content decoupling test returns a joint pass: per-class distinctness holds at all 10 of 10 sampled layers, global decoupling holds at 3 of 10 layers (L4, L8, L32, where pairwise cosine < 0.11), and the maximum pairwise cosine across layers is only 0.2349 (at layer 20). Low cross-class cosine plus high per-class AUC is the signature of separable, content-decoupled halt directions, and the halt signal remains separable from nuisance-variable baselines in the L20 to L24 band (Figure 2).

The *single global* halt direction is weaker and peaks elsewhere. Pooled global AUC peaks at layer 8 = 0.8571 and troughs at layer 32 = 0.7702, and at every layer the per-class AUCs (0.85 to 0.98) sit above the global AUC (0.77 to 0.86). The model does not carry one universal halt direction; it carries a family of class-conditioned ones.

4.2 Build versus read: the representation is assembled before it is consumed

Block-contribution decomposition separates where a halt representation is built from where it is read. For filled-cell-repeat (all documents, $N = 18$), the block-direction cosine is $+0.042$ at layer 20 versus -0.010 at layer 24; $+0.042$ at layer 20 is the largest single layer-class block contribution in the table. The positive value at layer 20 indicates the filled-cell halt direction is *built* there, while the negative value at layer 24 indicates layer 24 *reads* (consumes) rather than constructs it. This refines the class-specificity finding into a build-at-L20 / read-at-L24 offset for filled-cell-repeat, and positives ramp into and remain in a sustained high-halt-score regime at loop onset while controls stay flat (Figure 3).

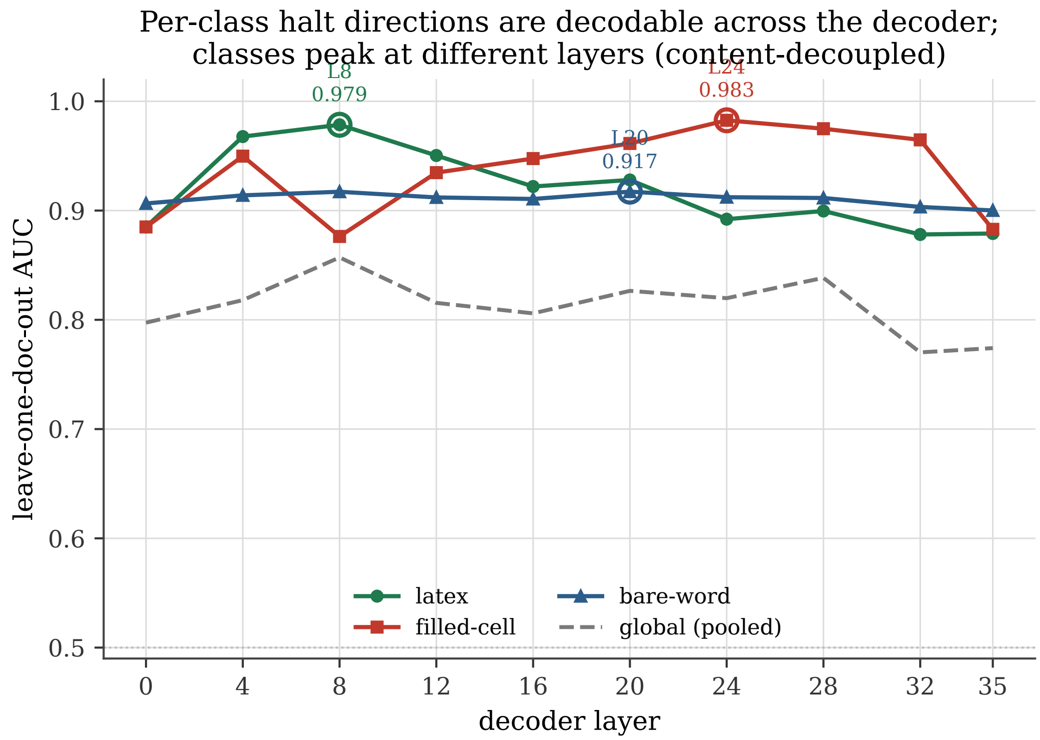

Figure 1: Per-class halt-direction leave-one-doc-out AUC across the decoder. The three surface classes peak at different layers (latex L8 = 0.979, filled-cell L24 = 0.983, bare-word L20 = 0.917), and per-class AUC (0.876 to 0.983) exceeds the pooled global AUC (0.770 to 0.857) at every layer. Distinct peak layers plus low cross-class cosine are the signature of separable, content-decoupled halt directions.

4.3 The downstream halt signal exists but arrives too late

Logit-lens analysis shows the network *does* compute a should-halt signal, just late. The EOS logit gap between positives and controls is most negative at layer 24 = −0.419, then flips positive to +0.481 at layer 32 and stays positive at +0.336 at layer 35 ($N = 4$ positives and 9 controls; Figure 4). The halt signal becomes legible at the unembedding only in the final layers, after the continuation state has been committed upstream.

4.4 EOS suppression is relative, not absolute

A naive reading would be that the model *crushes* EOS to a near-zero logit. The data say otherwise. Per-step analysis of a representative cap-hit document shows the chosen-token-minus-EOS raw logit margin has mean roughly 20.61 and median roughly 20.63 over the logged steps, with $P(\text{EOS})$ spanning roughly 10^{-8} to 10^{-10}. Strikingly, the cap-hit document's mean EOS logit (+10.76) is actually *higher* than a clean control's (+10.07). EOS is not absolutely suppressed; it is out-competed. Continuation tokens simply out-score it by about 20 logits, which is enough to crush the EOS probability into the 10^{-8} to 10^{-10} range while the EOS logit itself stays comfortably positive (Figure 5).

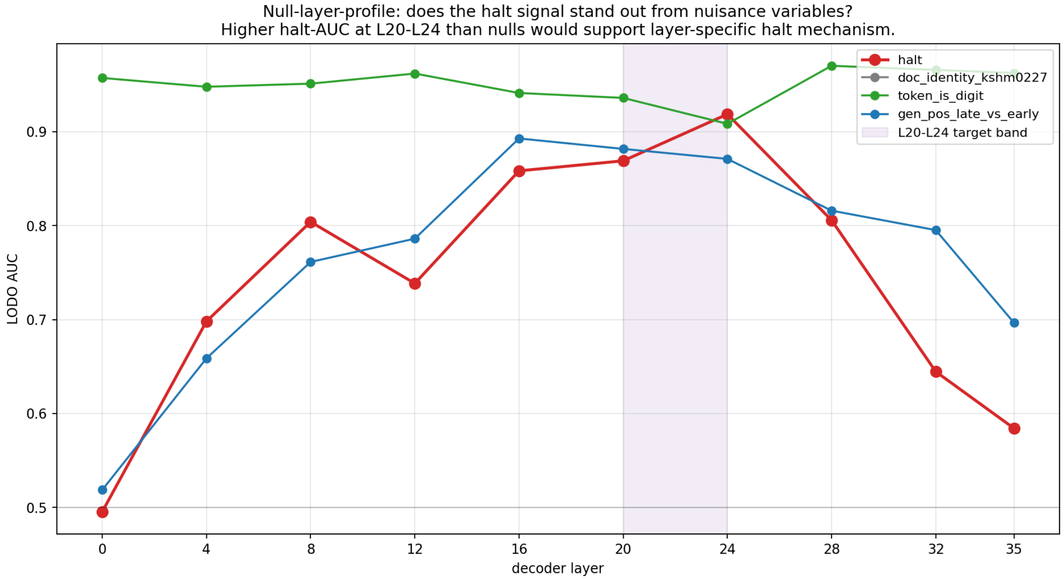

Figure 2: Null-layer profile: leave-one-doc-out AUC of the halt direction versus nuisance-variable probes (token-is-digit, generation-position late-vs-early, doc-identity) across layers. The halt signal peaks in the L20 to L24 band and is separable from the nuisance baselines there, supporting layer-localized halt detection over a trivial confound.

4.5 Calibration is refuted: EOS is genuinely missing at loop onset

If the failure were a near-miss in calibration, EOS would sit just below the argmax at loop onset. It does not. At loop-onset positions ($N = 110$ records over 22 documents), the EOS token sits at median rank roughly 10,502, and only 0.9% of records have EOS in the top 5. EOS is genuinely buried, not narrowly under-promoted. We note a framing caveat: programmatic n-gram loop-onset detection biases late, since records sit thousands of tokens deep, so "at loop onset" is a late-onset operationalization.

5 Causal tests: a converging series of nulls

The mechanism findings in Section 4 are correlationally rich. The natural follow-up is causal: can we *induce* or *abolish* halt-failure by perturbing a single direction or layer? Across six converging perturbation experiments — single-direction project-out, component-resolved patching, induction-head zeroing, prompt-engineering counterfactuals, SAE-feature ablation, and multi-layer coordinated patching, all returning the pre-registered FAIL/null verdict — the answer is no. We read this as *consistent with* a distributed attractor rather than a localized cause, but we flag at the outset that a converging set of nulls underdetermines the mechanism: it is equally consistent with a protocol that is under-powered at the halt-relevant locus, especially since three of the six rest on only $N = 2$ documents. We return to this disambiguation in Section 9; the present section reports the nulls and the one positive control that bounds, but does not eliminate, the under-powered alternative.

5.1 Sufficiency null at L0

Injecting the class-matched bare-word-repeat halt direction at layer 0 does not induce halt-failure. At the pre-registered target scale ($\alpha = 10$), the change in cap-hit rate versus the vanilla run is 0.0 percentage points, against a pre-registered PASS threshold of ≥ 40 pp, a clear failure of the sufficiency hypothesis. Layer 0 is a readout, not a generator.

5.2 Norm-scaled positive control bounds, but does not eliminate, "protocol under-powered"

A reviewer might object that the perturbation protocol is simply too weak to move anything. The norm-scaled positive control speaks to this, but only partially, and we state its composition precisely because it is load-bearing. The run scaled a single-direction layer-24 patch to 0.1×, 10×, and 100× the residual norm over four documents, but only *three* of those were cap-hit positives (manifest label `positive`: vanilla runs of 12,000 tokens). On those three positives, none escapes the cap at any scale — all stay pinned at 12,000. The perturbations nonetheless *do* land visibly in the output (one document gets a foreign-language token prepended, another gets stray Markdown emphasis inserted), so the protocol is demonstrably not inert: it delivers content-altering perturbations; it just cannot dislodge the halt-failure state. Content moves while token count does not. The fourth document (`table_N005_C04_s05000`, manifest label `control`) was *not* a cap-hit positive — it halts cleanly on its own at 477 tokens with an EOS stop — and so it cannot "escape" a cap it never hit. We flag it explicitly rather than fold it into the positive tally, because under the same perturbation its output *collapsed*: to 3 tokens at 10×, to 1 token at 0.1×, and to 12,000 tokens of degenerate single-character repetition at 100×. That collapse is a norm-shock signature; if anything it *weakens* the protocol-sensitivity rebuttal, by showing that this matched-norm patch can shatter a clean halt rather than cleanly perturb it, so we do not count it as a positive. We are careful about what the three positives rule out. They rebut the *strongest* form of the under-powered objection — that the protocol delivers no perturbation at all — but they do not rebut the weaker, more relevant form: that the protocol cannot perturb the *specific* halt-relevant direction at the *specific* halt-relevant locus. A random-direction patch that visibly reorders surface tokens (or shatters an unrelated clean halt) is not evidence that a targeted patch of the halt mechanism would have been detectable. We therefore treat the positive control as a sensitivity floor, not as a clean refutation of under-power.

5.3 The on-record null count, stated honestly

Our strongest necessity test is a reverse-direction project-out experiment: it asks whether *removing* the halt direction, by projecting it out of the residual stream at every position, lengthens generation. It does not. Across three clean-halting controls the projected-out run is unchanged (within 10%: +9.1%, 0%, 0%), and the positive document stays pinned at the cap, yielding a readout verdict. Importantly, the project-out hook is confirmed to fire: two of the four documents produce materially different output under it (one positive flips its table from Markdown to HTML), so this is a genuine null and not a silently inert intervention (Figure 6).

A component-resolved patch grid points the same way. It decomposes the patch into MLP, attention, and block contributions at layers 16, 20, and 24, across 5 documents and 4 intervention types: 0 of

45 real-patch cells produce any length reduction (a clean real-null), while 9 of 134 control cells do (Figure 7). The honest, pre-registered caveat is sharper than off-layer norm-shock alone: only 4 of the 9 nonzero controls are off-layer perturbations, while the remaining 5 fire *at the same layer and component as the real patch* (2 random-matched-norm, 3 shuffled-values). The clearest warning sign is at L16/block_output, where a random vector matched to the real patch's norm (≈ 47.4) reduced one document's length by 99.95% while the real halt-direction patch at exactly that site reduced it by 0%. This makes "the controls themselves are suspect" — the protocol's own perturbations can collapse generation in ways unrelated to any halt direction — a live alternative reading, not merely off-layer norm-shock, so the grid bounds protocol sensitivity rather than localizing a mechanism. The verified positive control above is our best evidence that the protocol has real sensitivity.

Accordingly, our load-bearing causal evidence is *six converging perturbation nulls*, all now reproducing from summary files in this repository and all returning the pre-registered FAIL/null verdict, with N ranging from 2 to 22 documents:

1. **B3, reverse-direction project-out null** (necessity; READOUT verdict; 3/3 controls unchanged, positive pinned at the cap, project-out hook confirmed to fire; `p2_pilot/b3_reverse_direction_summary.json`);
2. **P16, component-resolved patch grid** (necessity; FAIL verdict; 0/45 real-patch cells reduce length, 9/134 control cells do, $N = 5$ documents; `p16_component_resolved_short/_summary.json`). P16 subsumes the earlier component claims CL-19 and CL-20 and is counted once here, not as two separate nulls;
3. **P10b, induction-head zeroing** (necessity; FAIL verdict; 0/4 induction heads pass the pre-registered 30% threshold under zeroing, $N = 2$ documents; `p10b_head_zero_patching/_summary.json`);
4. **P19, system-prompt counterfactual** (FAIL verdict; 0/6 stop-instruction prompts pass, median reduction 0%, $N = 5$ documents; `p19_system_prompt_counterfactual/_summary.json`);
5. **P12, SAE-feature ablation** (FAIL verdict; 0/5 cross-validation folds pass all pre-registered criteria, top-feature AUC 0.54 to 0.63 against a 0.85 bar, $N = 22$ positives versus 56 controls; `p12_sae_ablation/_summary.json`);
6. **P22, multi-layer coordinated patch** (FAIL verdict; the real coordinated zero-patch of the L16/L20/L24 window reduces length 0%, $N = 2$ documents; `p22_multilayer_coordinated/_summary.json`).

These six are independent perturbation classes — project-out of a single direction, forward-patching of resolved components, head-level zeroing, prompt-level counterfactuals, SAE-feature ablation, and a multi-layer coordinated patch — and each carries its own positive or control evidence, so they are not a single test reported six times. We previously had only the first two reproducing from local data files and the other four recorded on cluster scratch; all six summary files are now synced and verified here, so we restate the count as *six converging perturbation nulls, all verified on disk.* We keep the count honest in the other direction too: three of the six rest on small samples ($N = 2$ for

B3's positive, P10b, and P22), and a uniform null at small N does not by itself separate a genuinely distributed mechanism from an under-powered protocol (Section 9). We also carry the controls-are-suspect caveat openly: in the component grid, 5 of the 9 nonzero controls fire at the same layer and component as the real patch (not merely off-layer), and a norm-matched random vector (≈ 47.4, the real patch's norm) reduced one document's length by 99.95% at L16/block_output where the real halt-direction patch reduced 0%. That a matched-norm random perturbation collapses generation where the real patch does nothing makes "the controls themselves are suspect" a live alternative to "the protocol is under-powered," rather than a simple off-layer norm-shock artifact.

We state the bottom line for this section plainly, because it is the load-bearing hedge of the paper. All single- and multi-site perturbations we ran returned null at this protocol's power. Whether the mechanism is genuinely distributed or the protocol is under-powered at the halt-relevant locus is *not resolved* by these data; the mechanism remains unidentified. The discriminating experiment — a positive-control patch of the *same* halt direction at a *non-loop* position, which would separate "distributed" from "under-powered" — is left to future work. The attractor reading in the rest of the paper is therefore a labeled hypothesis that fits the full evidence pattern, not a mechanism we have causally isolated.

6 Cross-family generality

The phenomenon reproduces beyond the single fine-tune, but the cross-family evidence is deliberately modest and we make no strong generality claim from it. The replication matrix is a 4×4 document-by-model grid spanning four model families (SmolDocling-256M-preview, InternVL3-8B, Qwen3-VL-8B-Instruct, and Qwen3-VL-30B-A3B-Instruct, a wide parameter range across multiple vendors) crossed with four documents, for sixteen cells in total. Under a strict criterion (`stop_reason == max_new_tokens`, i.e. the decode actually ran to the generation cap), halt-failure reproduces on *10 of the 16* document-by-model cells, with at least one cap-hit cell in every one of the four families and positive controls present. We stress that this is a small matrix: sixteen cells over four families is far too thin to support a generality claim, the per-family counts are 3/4, 1/4, 3/4, and 3/4 (InternVL3-8B reproduces in only one cell), and the directory-level aggregate verdict over this run abstains (`INSUFFICIENT_DATA`) rather than concluding generality. A more permissive near-cap relabel (counting a cell that stopped two tokens short of the cap) would raise the tally to 11/16, so even the count itself is criterion-dependent. We therefore read this only as the modest statement that halt-failure *reproduces on 10 of 16 doc-model cells across four families* while the aggregate verdict abstains, and we leave a properly powered strict-criterion cross-family study to future work.

Within the same vision-language family across checkpoints the picture is consistent: halt-failure reproduces across multiple checkpoints. As above, we present this cross-checkpoint reproduction qualitatively rather than with a precise cell count, which we defer to future work under a fixed strict stop-reason criterion.

7 Production fix: an honest precision-recall trade-off

A runtime monitor reading the layer-24 halt direction and boosting EOS can suppress runaway generation, but it sits on a genuine precision-recall trade-off rather than a free lunch.

At a +6.91 EOS boost, the reactive monitor achieves a 37.68% mean length reduction with 0 of 10 control false-positives and 5 of 13 positive cap-hits remaining (Figure 8). This is the clean-specificity operating point. At a +30 EOS boost, the same monitor achieves a 99.58% mean length reduction with 0 remaining cap-hits, but at a catastrophic 10 of 10 control false-positives. The two operating points bracket the trade-off: you can buy near-total loop suppression only by destroying clean-document specificity, so the deployable point is the conservative +6.91 boost.

Image content is sufficient to sustain the loop. Replacing the input image with grey noise collapses every tested cap-hit positive from a full-cap runaway to a short clean-EOS halt: across the tested positives, baseline cap-length stops become 13- to 18-token, EOS-terminated halts under grey-noise inpaint. All 5 of 5 explicitly named positives collapse this way (`arxiv_table_000266`, `docvqa_jqbg0227_p1`, `docvqa_kshm0227_p7`, `docvqa_srgb0228_p2`, `funsd_105_105`); with $N = 5$ we report the count qualitatively rather than attaching a Wilson interval. We state the claim as image *content* being sufficient to sustain the loop rather than as a clean vision-causality demonstration, because grey-noise replacement confounds removing visual evidence with destroying readable content; the discriminating experiment (swapping in a different readable document) is left to future work. This still motivates a vision-aware guardrail rather than a purely text-side penalty: the halt-failure state is conditioned on the visual input.

A correlational vision signal points the same way. In a single-positive-versus-single-control comparison, the positive document's text-to-image attention mass collapses in the L20 to L24 halt band: the positive-to-control ratio is 0.070 at layer 20 (14.2× lower), 0.216 at layer 24 (4.6× lower), and only a modest 0.695 at layer 16. We flag this as $N = 1$-vs-$N = 1$ and reserve it for future larger-sample work.

8 Related work

Our work sits at the intersection of seven threads: language-model degeneration, hallucination in vision-language models (VLMs), the mechanistic-interpretability toolkit, single-direction steering, the geometry of count-and-halt decisions, attention-sink accounts of termination, and cross-modal grounding collapse. We review each and locate our contribution against it.

Degeneration, repetition, and non-termination. Failure to terminate is a long-standing pathology of likelihood-trained autoregressive decoders: likelihood decoding produces bland, repetitive, degenerate text [14], and greedy, beam, top-k, and nucleus decoding are all provably inconsistent, able to emit infinite-length, zero-probability sequences unless the model is made self-terminating [40]. The training objective has itself been implicated, motivating unlikelihood training to suppress repetition at the source [39]. In production OCR pipelines the symptom is acute enough to be engineered around directly: retry mechanisms and post-hoc filters are now standard [29, 30], and repetition is documented as a first-class production concern in both general LLM serving [37] and recent VLM releases [38]. These works treat non-termination as a symptom to be suppressed at the decoding or serving layer rather than as a representational phenomenon to be localized. A complementary, mechanistically-flavored line asks *why* the decoder loops. The repeated-token phenomenon has been interpreted at the level of early-layer attention and sink structure [44], and single-neuron and single-head accounts of repetitive generation exist for text-only models: "repetition neurons" identify neuron-level circuits that drive repeats [13], while induction-head accounts trace pattern completion to match-and-copy heads [25] and quantify when those heads over-fire on repeated context [36] (note: this preprint was withdrawn in December 2025). These are the closest causal hypotheses to ours, and we situate against them deliberately. They are perpendicular on three axes: they study text-only LLMs rather than fine-tuned OCR-VLMs, they target repetition as such rather than the broader *halt-failure* phenomenon (of which token loops are one surface class among several, including phantom table rows and structural-marker hallucination), and they localize repetition to single neurons or heads (and, where intervened, report that ablating or regularizing them changes behavior). Our central finding is the opposite: a converging series of single-direction and single-layer perturbations *fails* to move the token count, which is why we read the failure as distributed rather than localized.

Hallucination in (V)LMs, including structural and OCR hallucination. Most VLM hallucination research concerns object hallucination in captioning and VQA, as captured by benchmarks such as POPE [20] and surveyed broadly in Bai et al. [5]. Document OCR introduces a distinct, *structural* failure mode: phantom blank table rows, fabricated page numbers and watermarks, and runaway markup loops; the closest mitigation-side work studies OCR hallucination under degraded input on the same model family with a dedicated benchmark and an uncertainty-aware fix [12]. We study this on a fine-tune of the Qwen2.5-VL family [4]; the same surface symptoms recur across OCR-VLM releases [15, 29, 30, 38]. The mechanistic substrate for OCR-specific behavior has begun to be mapped: text-inpaint causal protocols localize where OCR information enters the residual stream [32], and path-patching studies isolate vision-to-text "OCR heads" that route visual evidence into the language stream [3]. We borrow the text-inpaint logic for our image-content test, where replacing the input image with grey noise collapses tested cap-hit positives (all 5 of 5 named documents) from 12,000-token runaways into short `stop_reason=eos` halts of 13 to 18 tokens, establishing that image content is sufficient to sustain the halt-failure state (we say "sufficient" rather than "causally load-bearing" because grey noise confounds removing visual evidence with destroying readable content). Where these works localize *where OCR content arrives*, we localize *where the decision to terminate is represented* and show it forms a class-structured attractor rather than residing at a single routing site.

Mechanistic-interpretability methods. Our diagnostics are the standard mechinterp toolkit. We fit linear probes [1] for per-class halt directions across the decoder; the broader case that halt-relevant and quality-relevant signals are linearly decodable from VLM activations is consistent with prior probing work on the OCR setting. For causality we use activation and path patching in the lineage of causal-mediation and circuit-level analysis [22, 35], decomposed into MHSA, FFN, and hidden contributions via component-resolved causal tracing [19] so that we can separate where a halt representation is *built* from where it is *read*. We triangulate the depth at which a "should-halt" signal becomes legible at the unembedding using the logit lens [24] and its better-calibrated tuned-lens variant [6]; this is what surfaces our finding that the EOS logit gap is most negative at L24 and only flips positive by L32, i.e., the halt signal arrives too late to override an already-committed continuation state. Where a single probe direction is implicated, sparse-autoencoder decomposition [7, 8, 28] is the natural next step to ask whether it factors into interpretable sub-features, and we leave this to future work; the perturbations we do run are implemented as direct PyTorch forward hooks following the intervention-API conventions of [10, 41]. To ask whether the signature is introduced by fine-tuning we draw on the differential base-versus-fine-tune comparison vocabulary of feature crosscoders [23], and we run the comparison directly: the verified PCA crossover signature in the fine-tune (pos/ctl Cohen's-d ratio 3.79 / 2.74 / 2.78 at L16 / L20 / L24, all exceeding a pre-registered 2.0× threshold) is met, on the *same* Qwen2.5-VL-3B backbone before fine-tuning, by a flat profile (ratio 1.12 / 0.78 / 0.92, all near 1× and below the bar) computed over the base model's own cap-hit documents. The base backbone carries no separable halt geometry even where it runs to the cap, so fine-tuning is what introduces and sharply amplifies the structure that the attractor reading rests on. The "readout-versus-locus" distinction we draw at L24 shares its mechanism vocabulary with hallucination-detection work that reads model internals [33].

Single-direction and steering methods. The hypothesis that a behavior lives along a single residual-stream direction, testable by adding or projecting out that direction, is most cleanly stated for refusal, which is mediated by a single direction recoverable and steerable across many chat models [2], and has a direct method-class ancestor in inference-time intervention, which steers along an attention-head truthfulness direction [18]; the underlying add-a-difference-vector mechanism is the activation-engineering recipe our injections instantiate [34]. That single-direction picture is itself contested: refusal has been shown to factor into geometrically distinct, class-specific directions rather than one universal steerable axis [16], mirroring our class-specific halt directions. Our production monitor is the OCR-halt analogue: it reads an L24 halt direction and boosts the EOS logit rather than shifting an activation. The closest VLM-side comparable performs decoder-time steering pre-decoding rather than via a mid-decoder forward hook [31], and class-conditional steering offers a finer-grained alternative when one global direction is too coarse [43], relevant because our halt directions are class-specific, not universal. Crucially, our single-direction tests come back null. Injecting a class-matched halt direction at L0 changes the cap-hit rate by 0.0 percentage points against a pre-registered 40-point PASS threshold, and a norm-scaled positive control fails to dislodge the halt state at 0.1×, 10×, or 100× the residual norm (0 of 3 cap-hit positives escape the cap at any scale; a fourth, clean-halt control in the same run instead collapsed under the patch) even though the perturbations visibly alter output content. This is the inverse of the steering-success narrative in the single-direction literature: the direction is *decodable* (per-class AUC 0.876 to 0.983) without being *steerable*. We read this decodable-but-not-steerable gap as *consistent with* a distributed attractor, while noting it is equally consistent with a perturbation protocol that is under-powered at the halt-relevant locus; we do not adjudicate between the two (Section 9).

The counting-manifold account of halt decisions. The decision to stop is, in part, a decision about *how much* has been emitted, which connects halt-failure to the geometry of counting. The counting-manifold analysis reverse-engineers how a model tracks a running count along a low-dimensional curved manifold and uses it to make a stop-or-continue decision, with the count compared between two manifolds by rotation [11]. We take two things from it: the diagnostic prior to look for a manifold rather than a scalar register, and the warning that a single scalar threshold on a count-derived signal can be unstable. Our build-at-L20 / read-at-L24 offset for the filled-cell class (block-direction cosine $+0.042$ at L20 versus -0.010 at L24) is consistent with a representation that is assembled before it is consumed, in the spirit of that comparison-of-quantities picture, though we do not claim to have recovered a counting manifold specifically.

Attention sinks, EOS-halting, and SinkTrack. A prominent family of accounts ties termination and long-context stability to attention sinks. The attention-sink phenomenon was first identified on the initial tokens of a sequence and exploited for stable infinite-length streaming generation [42]. Building on it, SinkTrack treats an anchor token as an information sink and injects context into it to counteract attention drift, reporting multimodal gains [21], and the repeated-token failure has itself been traced to attention-sink structure [44]. This is the most actionable competing mechanism for our setting and a natural baseline. Our evidence is only partly aligned with it: we find that EOS suppression is *relative, not absolute*, with continuation tokens out-scoring EOS by roughly 20 logits while the EOS logit itself stays positive (a cap-hit document's mean EOS logit of $+10.76$ actually exceeds a clean control's $+10.07$), and that at loop onset EOS is genuinely buried (median rank near 10,500), rather than narrowly under-promoted by a single mis-set sink. A sink-anchoring intervention is therefore a complement to, not a substitute for, the distributed-attractor reading.

Cross-modal grounding collapse. Finally, several works frame VLM hallucination as a drift of attention away from visual evidence toward linguistic priors. Visual contrastive decoding contrasts outputs from original versus distorted images to suppress over-reliance on language priors [17], and a complementary line reorients attention toward image tokens at the moment grounding is needed: confidence-aware attention calibration rescales attention to counter late-generation modality drift [9], and attention-lens analyses diagnose text-vs-vision routing as the locus of modality bias and rebalance the two streams [45]. This "detect the decision moment, then intervene" structure matches our production setting, and the modality-bias account is corroborated on our side by a (single-positive-versus-single-control, hence preliminary) collapse of text-to-image attention mass in the L20 to L24 band.

Positioning. Against all of these, our contribution is to read OCR-VLM halt-failure not as a single broken layer, sink, head, or steerable direction, but *as if* it were a stable, class-structured attractor in the residual stream: decodable per class and content-decoupled, computed downstream too late to override an upstream continuation commitment, robust to single-site perturbation in both the necessity and sufficiency directions, fine-tune-introduced in its geometry, and conditioned on the visual input. We are explicit that "attractor" is a hypothesis we use to organize the evidence, not a mechanism we have causally localized: the converging nulls are equally consistent with an under-powered protocol, and we do not resolve that ambiguity here (Section 9). We treat the converging nulls as first-class evidence and report an honest production trade-off rather than a clean single-direction fix.

9 Limitations and walk-back

We state the walk-backs plainly because they are part of the contribution.

1. **The attractor reading is a hypothesis; "distributed" is not resolved against "under-powered."** This is the central walk-back. All single- and multi-site perturbations we ran returned null at this protocol's power, and whether the mechanism is genuinely distributed across many directions or the protocol is simply under-powered at the halt-relevant locus is not resolved by these data: the mechanism remains unidentified. The two readings are observationally equivalent at this protocol power, so we present "class-structured residual-stream attractor" as a labeled hypothesis and metaphor for the full evidence pattern, not as a causally-isolated mechanism. The experiment that would settle it — a positive-control patch of the same halt direction at a non-loop position — is left to future work. We make this hedge load-bearing throughout the title, abstract, and conclusion.

2. **Necessity evidence is six converging nulls, all now verified on disk but several at small N.** Six perturbation nulls now reproduce from summary files in this repository, all returning the pre-registered FAIL/null verdict: the reverse-direction project-out null (B3, readout verdict, controls unchanged, positive pinned at the cap, project-out hook confirmed to fire), the component-resolved patch grid (P16, 0 of 45 real-patch reductions; subsumes CL-19/CL-20, $N = 5$), induction-head zeroing (P10b, 0 of 4 induction heads pass under zeroing, $N = 2$), the system-prompt counterfactual (P19, 0 of 6 stop-instruction prompts pass, $N = 5$), SAE-feature ablation (P12, 0 of 5 folds pass all criteria, top-feature AUC 0.54 to 0.63 versus a 0.85 bar, $N = 22$ positives versus 56 controls), and a multi-layer coordinated zero-patch (P22, real coordinated patch reduces length 0%, $N = 2$). These were previously only partly synced; all six summary files are now on disk. We do not, however, let "six converging nulls" overstate the case. Three of the six rest on small samples (B3's positive document, P10b, and P22 each at $N = 2$), and a uniform null at $N = 2$ does not on its own discriminate a genuinely distributed mechanism from an under-powered protocol; that disambiguation (item 1 above) remains open. The residual caveat is sensitivity, not verification: in the P16 grid 5 of the 9 nonzero control patches fire at the same layer and component as the real patch (only 4 are off-layer), so the six nulls establish "single- and multi-site perturbation does not move halting at this protocol's power," not a fully exhaustive causal search.

3. **The controls are suspect, and the positive control only bounds under-power.** In the component grid, a norm-matched random vector (≈ 47.4) cut one document's length by 99.95% at L16/block_output where the real halt-direction patch cut 0%, and 5 of the 9 nonzero controls are not off-layer. That matched-norm random perturbations can collapse generation where the real patch does nothing means "the controls are suspect" is a real alternative to "the protocol is under-powered." Our verified positive control (Section 5) is the

load-bearing sensitivity evidence, but it rebuts only the strongest form of the under-powered objection (the protocol is not inert; it visibly alters content). It does not rebut the weaker, more relevant form, that the protocol cannot perturb the specific halt direction at the specific halt-relevant locus, which is why item 1 above remains open.

4. **Cross-family reproduction is small-matrix and the aggregate verdict abstains.** Under the strict `stop_reason == max_new_tokens` criterion, halt-failure reproduces on 10 of 16 doc-model cells across four families (per-family 3/4, 1/4, 3/4, 3/4), but the matrix is only sixteen cells and the directory-level aggregate verdict over this run is `INSUFFICIENT_DATA`. We make no strong generality claim from it. A permissive near-cap relabel would shift the tally to 11/16, so the count is criterion-dependent. A properly powered strict-criterion cross-family (and cross-checkpoint) study is left to future work.

5. **Fine-tune origin: both halves of the comparison are now verified.** In the fine-tuned model the positive-versus-control absolute Cohen's-d crossover ratio at layers 16, 20, and 24 is 3.79, 2.74, and 2.78 ($N = 22$ positives versus $N = 52$ controls), all exceeding the pre-registered 2.0$\times$ threshold. The matching base-model comparison, previously deferred, has now been run on the same Qwen2.5-VL-3B backbone before fine-tuning and is persisted on disk: over the base model's own cap-hit documents ($N = 10$ base cap-hits versus $N = 15$ controls) the ratio triple is 1.12, 0.78, and 0.92, all near 1$\times$ and all below the 2.0$\times$ bar. The base backbone therefore shows no separable halt geometry even on the documents where it itself runs to the cap, while the fine-tune exceeds the bar at every layer. We read this as direct evidence that fine-tuning introduces and sharply amplifies the separable geometry, and we no longer treat the base comparison as a missing run. The residual caveat is one of scale: the base comparison rests on a 10-versus-15 archetype split rather than the fine-tune's 22-versus-52.

6. **The vision-attention-collapse signal is $N = 1$-vs-$N = 1$.** The attention-collapse comparison is a single-positive-versus-single-control measurement, not yet statistically meaningful, and is reserved for future larger-sample work.

7. **Loop-onset framing biases late.** The calibration result is computed at programmatically detected onset positions that sit thousands of tokens deep; this overstates "onset" and is a framing concern, though it does not change the numeric finding that EOS is genuinely missing.

10 Conclusion

Halt-failure in a fine-tuned OCR-VLM *behaves like* a stable, class-structured attractor in the residual stream rather than a single broken layer. We advance this as a labeled hypothesis, not a settled mechanism. The evidence is pre-registered and converging: per-class halt directions are linearly decodable (AUC 0.876 to 0.983) and content-decoupled (max pairwise cosine 0.2349); the network computes a downstream should-halt signal that arrives too late to override an upstream continuation state (logit-lens EOS gap -0.419 at layer 24 flipping to $+0.481$ by layer 32), and that suppression is relative not absolute (chosen-vs-EOS margin about 20 logits; cap-hit EOS logit $+10.76$ exceeding a control's $+10.07$); EOS is genuinely missing at loop onset (median rank roughly

10,502); and six converging single-direction, single-layer, and multi-layer perturbations fail to move token count. We are deliberately careful with the last point: all single- and multi-site perturbations we ran returned null at this protocol's power, and whether the mechanism is genuinely distributed or the protocol is under-powered at the halt-relevant locus is *not resolved* by these data — the mechanism remains unidentified. The verified norm-scaled positive control (0 of 3 cap-hit positives escape at any scale, content visibly changing; a fourth clean-halt control in the run instead collapsed under the same patch, a norm-shock signature that blunts rather than strengthens the rebuttal) rebuts only the strongest, protocol-is-inert form of the under-powered objection; it does not rebut the weaker form, that the protocol cannot perturb the halt-relevant direction at the halt-relevant locus. Our load-bearing causal evidence is six converging perturbation nulls, all now reproducing from summary files in this repository and all returning the pre-registered FAIL/null verdict (N ranging 2 to 22 documents): the B3 reverse-direction project-out, the P16 component-resolved grid (which subsumes CL-19/CL-20), P10b induction-head zeroing, the P19 system-prompt counterfactual, P12 SAE-feature ablation, and the P22 multi-layer coordinated patch. We are careful not to let the count overstate the case: three of the six rest on $N = 2$ (B3's positive, P10b, P22), and a uniform null at small N does not by itself separate "distributed" from "under-powered." A matched base-model comparison on the same Qwen2.5-VL-3B backbone substantiates the fine-tune-origin half of the reading: the base model carries no separable halt geometry even on its own cap-hit documents (pos/ctl Cohen's-d ratio 1.12 / 0.78 / 0.92 at L16 / L20 / L24, all near $1\times$ and below the $2.0\times$ bar) whereas the fine-tune scores 3.79 / 2.74 / 2.78, so fine-tuning is what introduces and sharply amplifies the structure. The surface symptom is not unique to the studied fine-tune: under a strict stop-reason criterion halt-failure reproduces on 10 of 16 doc-model cells across four model families, but the matrix is small and the directory-level aggregate verdict abstains, so we draw no strong generality claim from it. We present a deployable production monitor with an explicit precision-recall trade-off (37.68% reduction at zero control false-positives, or 99.58% at 10 of 10 false-positives) plus an image-inpaint result showing that image content is sufficient to sustain the loop (all 5 of 5 named positives collapse to short EOS halts under grey noise). The walk-backs are part of the result: showing where the single-cause story fails is what motivates — without proving — the distributed attractor reading, whose fine-tune-introduced geometry is now supported by the base comparison, and the experiment that would distinguish "distributed" from "under-powered" is the clearest next step.

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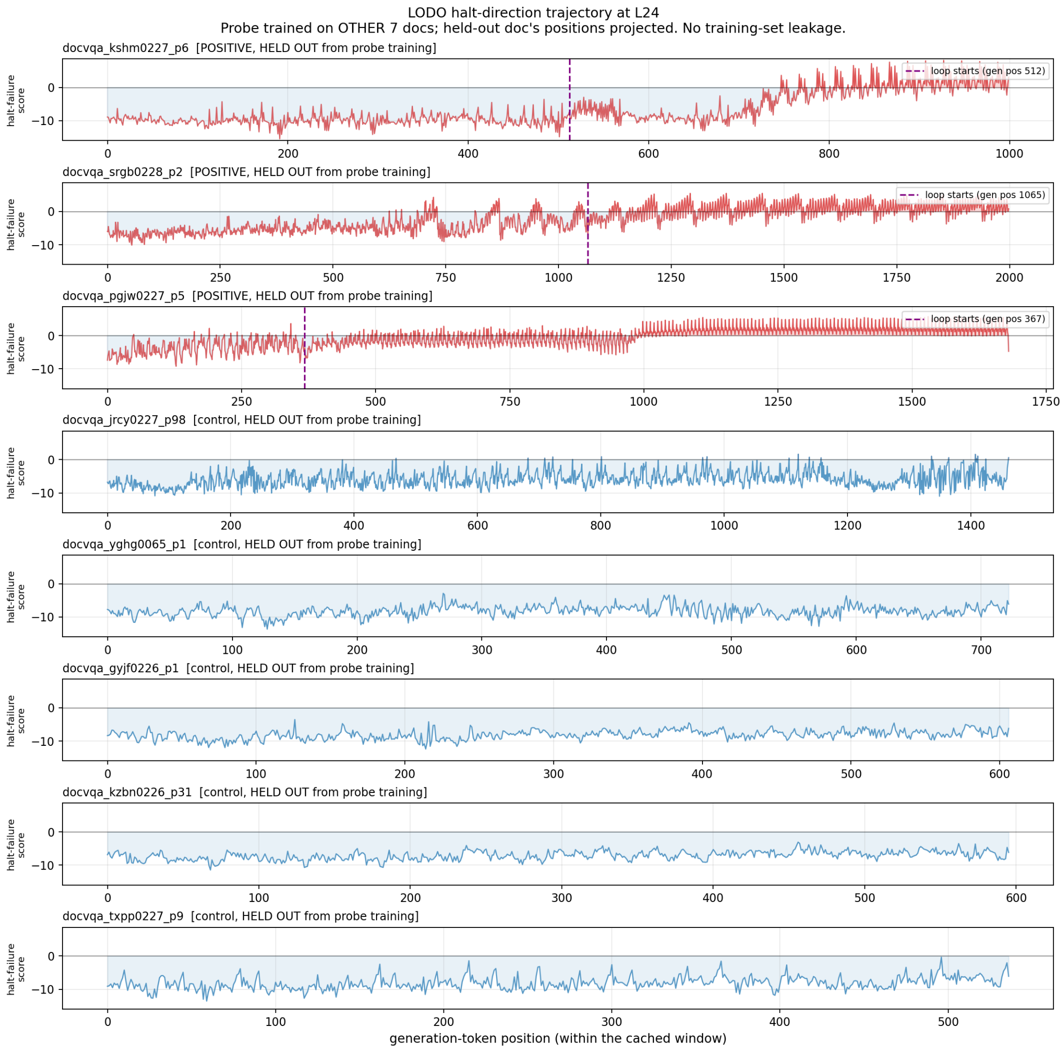

Figure 3: Per-position projection onto the L24 halt direction over the decode, under leave-one-doc-out (held-out docs only, no probe training-set leakage). The three positives (top) ramp into a sustained high-halt-score regime at loop onset, while the five controls (bottom) stay flat at baseline. Visualizes the attractor reading: positives enter and remain in a distinct residual-stream region.

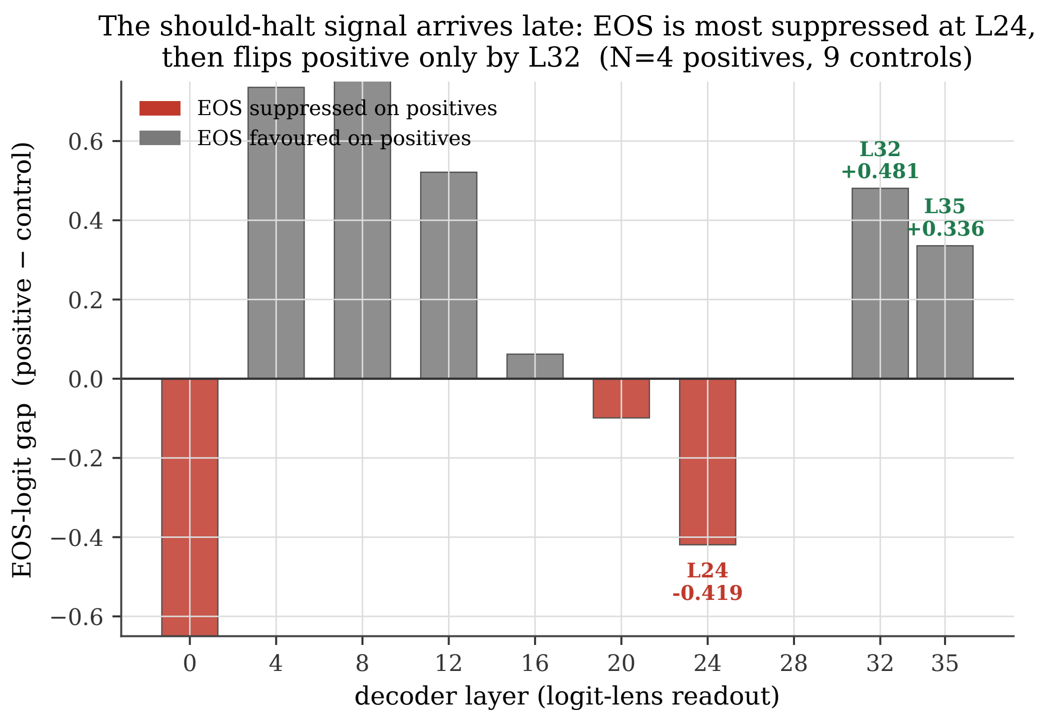

Figure 4: Logit-lens EOS-logit gap (positive minus control) by decoder layer. The gap is most negative at L24 (-0.419), meaning EOS is most suppressed there, then flips positive to $+0.481$ at L32 and $+0.336$ at L35. The should-halt signal becomes legible at the unembedding only in the final layers, after the continuation state is committed upstream ($N = 4$ positives, 9 controls).

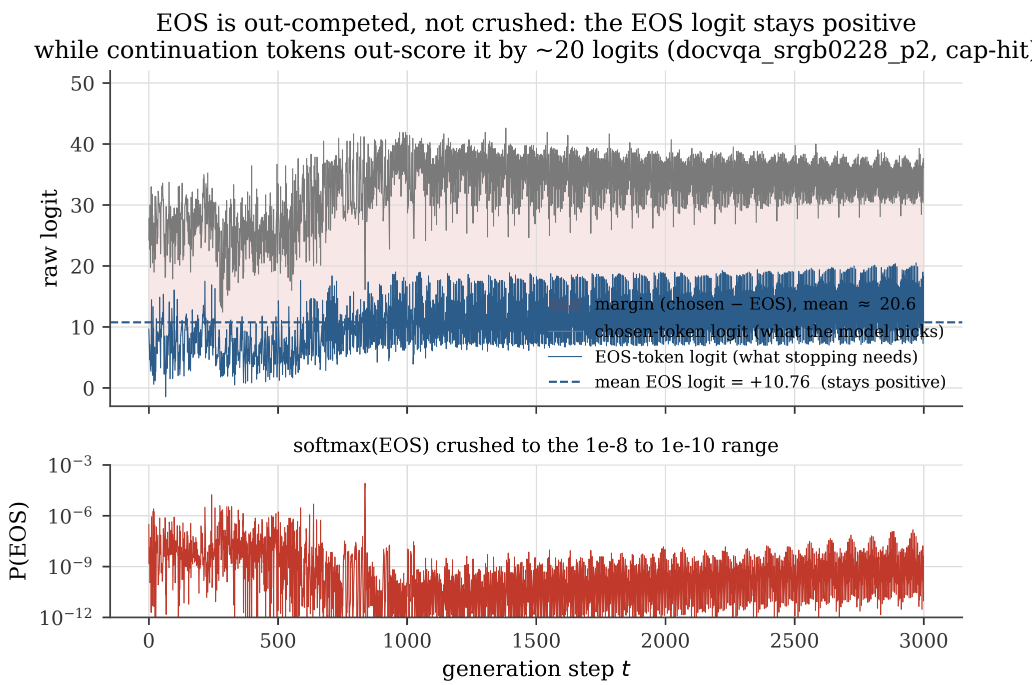

Figure 5: EOS suppression is relative, not absolute (cap-hit document). The EOS logit stays positive throughout (mean +10.76, which actually exceeds a clean control's +10.07), while continuation tokens out-score it by a mean margin of roughly 20.6 logits, crushing softmax(EOS) into the 10^{-8} to 10^{-10} range.

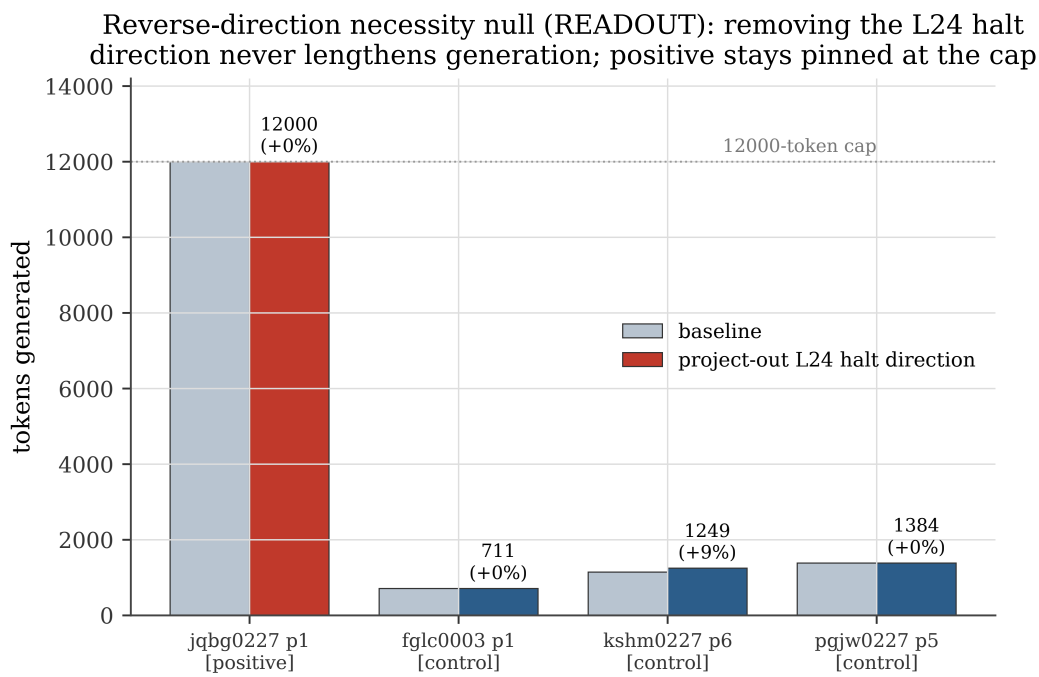

Figure 6: Reverse-direction necessity null (READOUT verdict). Projecting out the L24 halt direction at every position does not lengthen generation: the positive document stays pinned at the 12000-token cap (12000 to 12000, +0%) and the three clean-halting controls are unchanged (+9.1%, 0%, 0%). The project-out hook is confirmed to fire (output content changes without length change), so this is a genuine null.

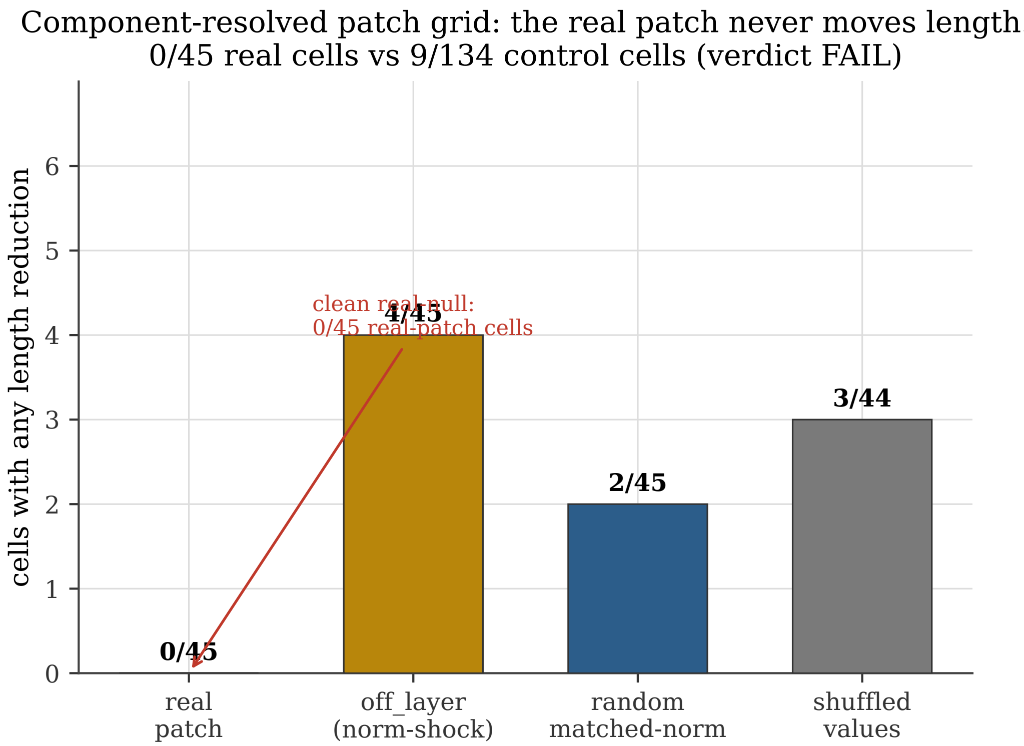

Figure 7: Component-resolved patch grid by intervention type. The real patch never moves length (0/45 cells, a clean real-null), while 9/134 control cells show a nonzero reduction: 4 off-layer, 2 random-matched-norm, and 3 shuffled-values. Only 4 are off-layer; the other 5 fire at the same layer and component as the real patch. Most starkly, at L16/block_output a random vector matched to the real patch's norm (≈ 47.4) cuts one document's length by 99.95% while the real patch there cuts 0%. Verdict FAIL. These nonzero controls make "the controls are suspect" a live reading and bound protocol sensitivity rather than localizing a mechanism.

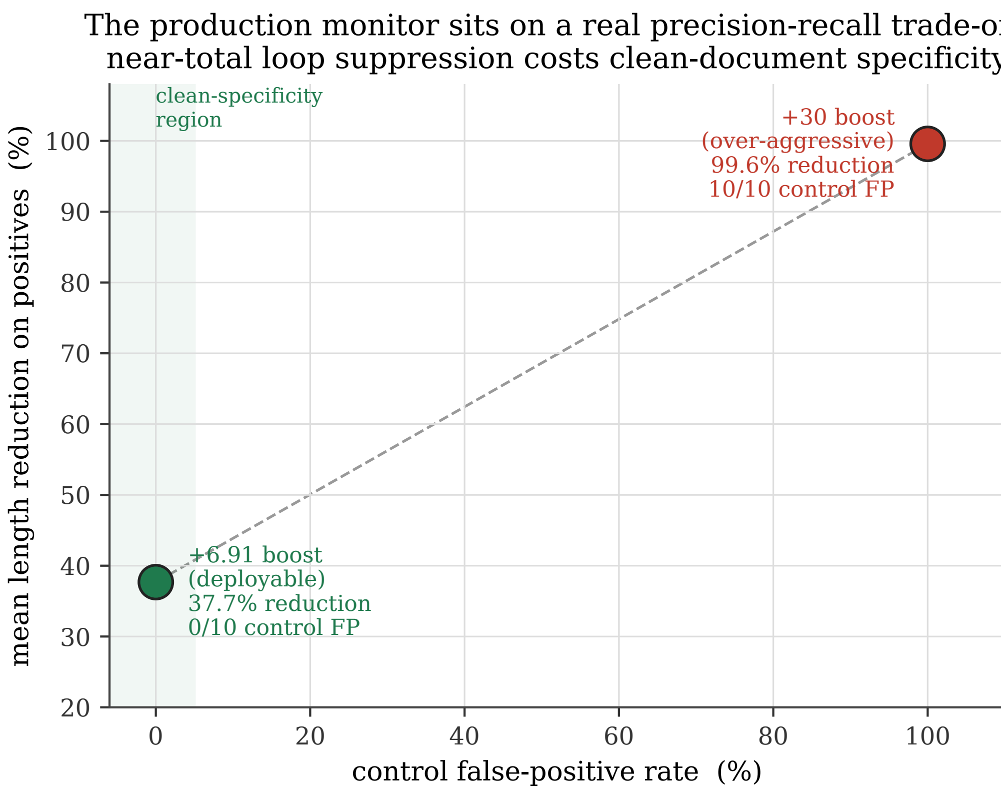

Figure 8: The production monitor sits on a real precision-recall trade-off. At a +6.91 EOS boost it gives 37.7% mean length reduction with 0/10 control false-positives (deployable clean-specificity point); at a +30 boost it gives 99.6% reduction with 0 remaining cap-hits but 10/10 control false-positives. Near-total loop suppression costs clean-document specificity. $N = 13$ positives, 10 controls.